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A Supplementary visualization

A.1 Wave-like patterns in embeddings

Figure 2 displays the sinusoidal patterns in Llama 3 70B and OLMo2 13B after PCA dimensionality reduction. For clarity, we only include the first 16 principal components.

A.2 Explainability plots for arithmetic tasks.

Model behavior. To better explain the behavior of the LMs, we conduct a simple circuit analysis and a feature attribution experiment using integrated gradients (Sundararajan et al., 2017). For convenience, we focus on the two smaller models in our panel. OLMo 2 1B and Llama 3 1B.

Both experiments suggest one major difference between operand pairs leading to failure and to success: the probability assigned by the LLM to the predicted output token tends to be statistically lower when the model produces an incorrect output, see Figure 4. We also observe the same subset of heads being activated for failure and success on the arithmetic task. Besides the usefulness of this difference in probability mass for diagnostic purposes, these experiments also suggest a difference in *degree* rather than *kind* between failures and successes.

In Figure 5, we present an overview of head-level attribution of the logits in Llama 2 1B. The same heads in Layers 13 through 15 appear activated in all cases, playing the same inhibitor and booster roles. Incorrectly performed addition leads to a noisier overall pattern. Remarkably, we observe that activity occurs in the latter stages of the model, whereas input embeddings (layer 0) already contain precise numeric information, as per our probing experiments. This delayed processing may explain some of the errors we observe, despite the high accuracy of our probes in Section 3.

A.3 Analysis of sin-base probes' learned representations

In Figure 6, we can see that our newly proposed sin-like probes indeed learn to project input embeddings of models into an expected, generalized wave-like representation.

B Experimental details

1	"x_1+x_2 equals to "
2	"The result of x_1+x_2 is "
3	"The result of x_1 plus x_2 is "
4	"The result of x_1 plus x_2 = "
5	"The result of x_1 plus x_2 ="
6	"x_1 plus x_2 equals to "
7	"x_1+x_2="
8	"x_1 plus x_2 equals "
9	"x_1 plus x_2 is equal to "
10	"x_1+x_2 equals "
11	"x_1+x_2 is equal to "
12	"x_1 plus x_2 equals "
13	"x_1 plus x_2 is equal to "

(a) Prompts considered for addition task. x_1 and x_2 are placeholders for the augend and the addend. Prompts are delimited by double quotes; trailing white-space is significant.

1	"The result of x_1 minus x_2 is "
2	"The result of x_1 minus x_2 = "
3	"The result of x_1 minus x_2 ="
4	"x_1 minus x_2 equals to "
5	"x_1-x_2="
6	"x_1 minus x_2 equals "
7	"x_1 minus x_2 is equal to "
8	"x_1-x_2 equals "
9	"x_1-x_2 is equal to "
10	"x_1 minus x_2 equals "
11	"x_1 minus x_2 is equal to "

(b) Prompts considered for subtraction task. x_1 and x_2 are placeholders for the minuend and the subtrahend. Prompts are delimited by double quotes; trailing white-space is significant.

Table 3: Prompts considered for engineering of arithmetic zero-shot setting.

We conduct a minimal prompt optimization in Section 4 to maximize the performances on arithmetic task. For all models below 20B parameters, we explore the prompts listed in Table 3a and Table 3b, and report results with the highest performance in a zero-shot setting in Section 4.

The most successful prompts for addition are prompt #4 for Llama 3 1B, 2B and 8B as well as OLMo2 1B, and prompt #3 for OLMO2 7B and 13B. As for subtraction, the most effective prompt was prompt #1 for Llama 3 1B, OLMo2 1B and 7B, and prompt #2 for Llama 3 3B and 8B as well as OLMo2 13B.

C Disclosure of usage of AI assistance

We disclose that we used AI assistance during implementation of this work and its writing. Specifically, we used AI-based code auto-completion (Github Copilot) for increasing productivity of programming, and conversational chatbots (OpenAI ChatGPT, Google Gemini) for improving grammar and fluency of the text. We guarantee that all content is original and factually accurate.

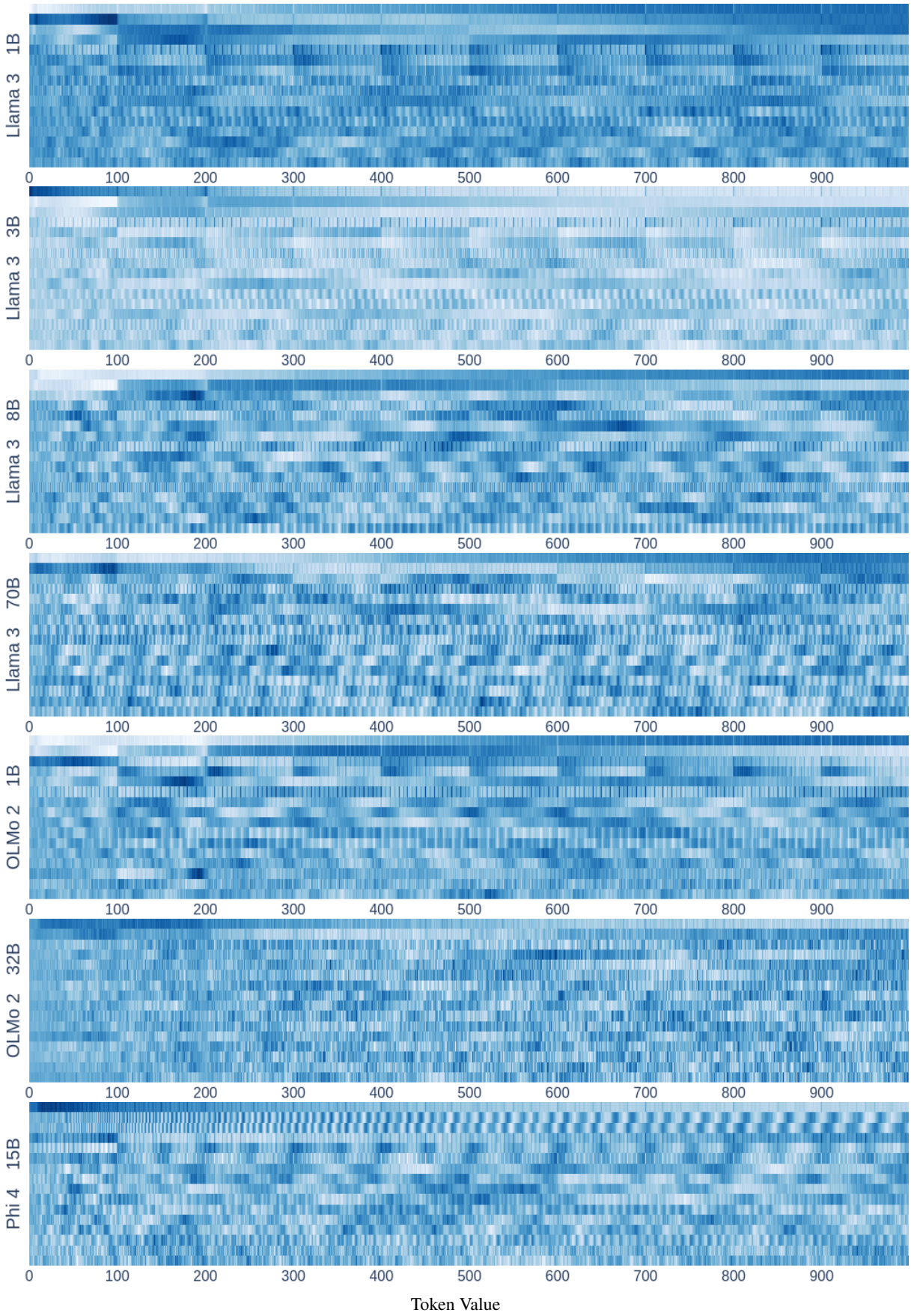

Figure 2: Visualization of PCA (DIM=16) reduced number embeddings, selected models. Although most model exhibit relatively regular wave-like patterns, OLMO 2 32B exhibit little regularity.